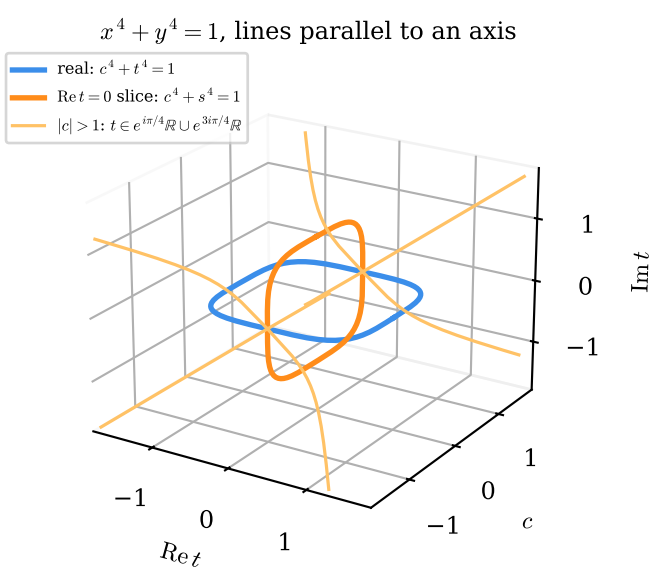
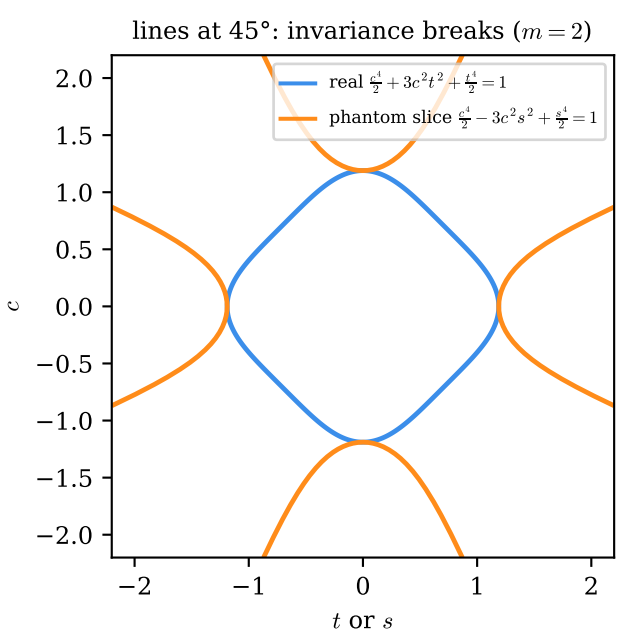

$x^4 + y^4 = 1$, lines parallel to an axis
real: $c^4 + t^4 = 1$
$\mathrm{Re}\, t = 0$ slice: $c^4 + s^4 = 1$
$|c| > 1$: $t \in e^{i\pi/4}\mathbb{R} \cup e^{3i\pi/4}\mathbb{R}$
$\mathrm{Im}\, t$
$\mathrm{Re}\, t$
c


lines at 45°: invariance breaks ($m = 2$)
real $\frac{c^4}{2} + 3c^2t^2 + \frac{t^4}{2} = 1$
phantom slice $\frac{c^4}{2} - 3c^2s^2 + \frac{s^4}{2} = 1$
c
t or s